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# AI PRACTICAL 4
# Information Retrieval System using NLP (NLTK)
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# STEP 1 : INSTALL NLTK
# Run only first time

# pip install nltk


# STEP 2 : IMPORT LIBRARIES

import nltk

from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords

from collections import Counter

from nltk import pos_tag


# STEP 3 : DOWNLOAD NLTK DATA
# Run only first time

nltk.download('punkt')

nltk.download('stopwords')

nltk.download('averaged_perceptron_tagger')


# STEP 4 : INPUT TEXT

text = "Artificial Intelligence is changing the world with smart
technology"

print("Original Text:\n")

print(text)


# =====
# a) TEXT TOKENIZATION
# =====

tokens = word_tokenize(text)

print("\nTokenized Words:\n")

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print(tokens)

# =====
# b) COUNT WORD FREQUENCY
# =====

frequency = Counter(tokens)

print("\nWord Frequency:\n")

print(frequency)

# =====
# c) REMOVE STOP WORDS
# =====

stop_words = set(stopwords.words('english'))

filtered_words = []

for word in tokens:

    if word.lower() not in stop_words:

        filtered_words.append(word)

print("\nAfter Removing Stop Words:\n")

print(filtered_words)

# =====
# d) POS TAGGING
# =====

pos = pos_tag(tokens)

print("\nPOS Tagging:\n")

print(pos)

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